Mid-Sem 2025

① (a) $\frac{dC}{dt} = \dot{\varepsilon} p_\eta + \varepsilon \dot{p}_\eta - \eta \dot{p}_\varepsilon - \dot{\eta} p_\varepsilon$

Now, $p_\eta = \frac{\partial L}{\partial \dot{\eta}} \Rightarrow \dot{p}_\eta = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\eta}}\right) = \frac{\partial L}{\partial \eta}$

$p_\varepsilon = \frac{\partial L}{\partial \dot{\varepsilon}} \Rightarrow \dot{p}_\varepsilon = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\varepsilon}}\right) = \frac{\partial L}{\partial \varepsilon}$

from Langrange's Eqn

$\therefore \frac{dC}{dt} = \dot{\varepsilon}\frac{\partial L}{\partial \varepsilon} = m\varepsilon(\dot{\varepsilon}^2 + \dot{\eta}^2) - \frac{4k\varepsilon}{(\eta^2+\varepsilon^2)^2} = \dot{p}_\varepsilon$

$\frac{\partial L}{\partial \eta} = m\eta(\dot{\varepsilon}^2 + \dot{\eta}^2) - \frac{4k\eta}{(\eta^2+\varepsilon^2)^2} = \dot{p}_\eta$ (from symmetry)

$\therefore \frac{\partial L}{\partial \dot{\eta}} = m\dot{\eta}(\varepsilon^2 + \eta^2) = p_\eta$

$\frac{\partial L}{\partial \dot{\varepsilon}} = m\dot{\varepsilon}(\varepsilon^2 + \eta^2) = p_\varepsilon$

$\therefore \frac{dC}{dt} = \dot{\varepsilon} m\dot{\eta}(\varepsilon^2 + \eta^2) + \varepsilon\left[m\eta(\dot{\varepsilon}^2 + \dot{\eta}^2) - \frac{4k\eta}{(\eta^2+\varepsilon^2)^2}\right]$

$- \eta\left[m\varepsilon(\dot{\varepsilon}^2 + \dot{\eta}^2) - \frac{4k\varepsilon}{(\eta^2+\varepsilon^2)^2}\right] - m\dot{\eta}\dot{\varepsilon}(\varepsilon^2 + \eta^2)$

$\therefore \boxed{\frac{dC}{dt} = 0}$

$\therefore$ ~~C is a con~~

$\therefore$ C is a constant of motion.

(ii) L does not have any explicit time-dependence. Hence total energy E is conserved.

$$T = \frac{m}{2}(\xi^2+\eta^2)(\dot{\xi}^2+\dot{\eta}^2)$$

$$V = \frac{2k}{\xi^2+\eta^2}$$

(iii) Yes. 2 GCs (ξ, η) & 2 constants of motion.

(a) E (b) C

② (i) $x = r\cos\phi$; $y = r\sin\phi$.

$\dot{x} = \dot{r}\cos\phi - r\sin\phi\,\dot{\phi}$; $\dot{y} = \dot{r}\sin\phi + r\cos\phi\,\dot{\phi}$

$$v^2 = \dot{x}^2+\dot{y}^2 = \dot{r}^2 + r^2\dot{\phi}^2$$

$$\therefore T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2)$$

$$\therefore V = -\frac{k}{r} + \frac{\beta}{2r^2}$$

$$\therefore L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) + \frac{k}{r} - \frac{\beta}{2r^2}$$

For ϕ:

$$\frac{\partial L}{\partial \phi} = 0 \quad ; \quad \frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi}$$

$$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = 0 \Rightarrow mr^2\dot{\phi} = \text{constant} = L$$

$$\Rightarrow \boxed{\dot{\phi} = \frac{h}{r^2}} \quad \left[h = \frac{L}{m}\right] \quad \text{— (1)}$$

For r: $\frac{\partial L}{\partial \dot{r}} = m\dot{r}$

$$\frac{\partial L}{\partial r} = mr\dot{\phi}^2 - \frac{k}{r^2} + \frac{\beta}{r^3}$$

$$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0$$

$$\Rightarrow m\ddot{r} - mr\left(\frac{h}{r^2}\right)^2 + \frac{k}{r^2} - \frac{\beta}{r^3} = 0.$$

$$\Rightarrow m\ddot{r} - \frac{mh^2}{r^3} + \frac{k}{r^2} - \frac{\beta}{r^3} = 0.$$

$$\Rightarrow \boxed{m\ddot{r} + \frac{k}{r^2} - \frac{1}{r^3}(mh^2 + \beta) = 0} \text{ — EOM}$$

(iii) $u(\phi) = \frac{1}{r} \Rightarrow r = \frac{1}{u(\phi)}$.

$$\therefore \dot{r} = \frac{dr}{dt} = \frac{dr}{du} \cdot \frac{du}{d\phi} \cdot \frac{d\phi}{dt}.$$

$$= -\frac{1}{u^2}\dot{\phi}\frac{du}{d\phi} = -h\frac{du}{d\phi} \quad \text{(from (1))}$$

$$\therefore \ddot{r} = \frac{d\dot{r}}{dt} = \frac{d}{dt} -h\frac{d^2u}{d\phi^2} \cdot \frac{d\phi}{dt}$$

$$= -h\dot{\phi}\frac{d^2u}{d\phi^2} = -h^2u^2\frac{d^2u}{d\phi^2} \quad \text{(from (10))}$$

Replacing these in EOM:

$$-mh^2u^2\frac{d^2u}{d\phi^2} + ku^2 - u^3(mh^2 + \beta) = 0$$

$$\Rightarrow \boxed{\frac{d^2u}{d\phi^2} + u\left(\cancel{mh^2}\ 1 + \frac{\beta}{mh^2}\right) = \frac{k}{mh^2}} \text{ — Orbital Eq}^n$$

(1) (a) Total energy E — L is independent of time

(b) $\frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi}$ = angular momentum.